Hack-Nation

In collaboration with MIT Club of Northern California and MIT Club of Germany

CHALLENGE

01

Generative City-Wallet

Hyperpersonalized offers for anyone, anywhere

POWERED BY

DSV Gruppe

DSV-Gruppe — Deutscher Sparkassenverlag
(A Company of the German Savings Banks Financial Group)

Goals and Motivation

Meet Mia

Mia is 28, works in marketing, and is walking through Stuttgart's old town on a Tuesday lunch break — twelve minutes to spare, slightly cold, vaguely hungry, phone in hand.

The system knows quite a lot about this moment. It knows it is 11°C and overcast. It knows there is a café 80 metres away that has been quiet all morning and has just brewed a fresh batch. It knows Mia has stopped twice in the last ten minutes and is moving slowly — the behavioural signature of someone browsing, not commuting. It knows she has responded to warm-drink offers before.

None of this is being used.

Instead, Mia's phone shows her a push notification from a coupon app: 10% off at a restaurant she has never visited, valid for the next thirty days.

This is not a technology problem. The data exists. The location is precise. The café's quiet period is measurable. Mia's likely intent is inferable. What is missing is the layer that connects these signals in real time and turns them into a single, specific, well-timed offer — not a generic discount, but *this café, this drink, right now, because the moment is right.*

That gap — between a person and a perfectly relevant local offer that already exists two minutes from where they are standing — is what this challenge asks you to close.

Current challenges

Personalization is a tool often not deployed by smaller local shops. Build the app that gives DSV Gruppe's local merchant partners access to the personalisation infrastructure they have never had — and use it to make Mia's next fifteen minutes count.

STATIC OFFERS	Traditional loyalty programmes and coupon books are static and often irrelevant. A 10% discount valid for a month does almost nothing to drive a spontaneous visit today.
NO ALGORITHMIC POWER	Small local retailers lack the marketing resources and data analysts to compete with the dynamic pricing models and recommendation algorithms of global e-commerce giants. The infrastructure exists but dynamic content is missing.
CONTEXT BLINDNESS	Even where offers and users are in the same street at the same moment, there is no connective tissue. The gap between a slow afternoon at a café and a person looking for a warm place to sit is never closed.

Your Challenge

Build CITY WALLET — a working end-to-end MVP for an AI-powered city wallet that detects the most relevant local offer for a user in real time, generates it dynamically, and makes it redeemable through a simulated checkout.

At the centre is a mobile user experience that surfaces locally relevant offers in everyday situations — not as static coupons but as dynamically generated, context-aware recommendations. These are

grounded in real-time signals: weather, time of day, location, local events, and demand patterns. The solution serves end users first, but also connects local merchants: merchants participate with minimal

effort by setting simple rules or goals, while the AI generates the actual offer automatically (within guiding parameters).

UX design is as much part of the challenge as the technology. How, when and in what form the offer appears determines whether it is accepted or ignored.

A Living Wallet — not a coupon app

Your solution must work as a real-time context layer — not a static database of offers. Think of it as a system where offers don't exist until the moment they are needed, and are generated specifically for this user, this location, this minute. The merchant sets rules and goals; the AI creates the offer. Country-specific or city-specific parameters (merchant inventory signals, local event data, weather, time patterns) should be inputs to your system, not hardcoded assumptions.

Build all three of these modules:

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01 Context Sensing Layer

Aggregate real-time context signals: weather data, local event calendars (city festivals, sports events), user location via geo-fencing, and — as a key DSV asset — Payone transaction density at nearby merchants. The system must recognise a composite context state (e.g. 'raining + Tuesday afternoon + partner café transaction volume unusually low') and trigger the generative pipeline. Context signals must be configurable without changing the codebase — a different city or data source should slot in as a configuration, not a rewrite.

Required: *Must incorporate at least two real context signal categories visible to the user — weather, location, time, local events or demand proxies.*

02 Generative Offer Engine

Based on the context state, the system autonomously generates a targeted campaign: content, discount parameters, visual design, and timing. This is not template-filling — use Generative UI (GenUI) techniques to create a fitting interface element (widget) including appropriate imagery, tone, and emotional framing. The merchant specifies only rules or goals ('max 20% discount to fill quiet hours'); the AI handles the creative execution. On-device Small Language Models (SLMs) are encouraged for privacy compliance (GDPR): local preference and movement data should not reach the cloud. Only an abstract 'intent' signal is sent upstream.

Required: *Offer must be generated dynamically, not retrieved from a static database. Show the merchant-side rule interface — even as a mockup.*

03 Seamless Checkout & Redemption

When the user accepts an offer, the system generates a dynamic QR code or token that is validated via API. The redemption experience must be seamless to the point of a simulated checkout: QR scan, token, or cashback mechanic. Alternatively — and more simply — after a successful transaction the customer receives a cashback credit for the discount amount. Build both a consumer view and a merchant view: the merchant sees offer performance and accept/decline rates in aggregate.

Required: *Must demonstrate end-to-end flow from offer generation to simulated redemption. Merchant dashboard or summary view required, even as static mockup.*

The UX Requirement

Design is not decoration; it is the mechanism of acceptance or rejection. Your prototype must address the following explicitly:

- Where does the interaction happen? (Push notification, in-app card, lock-screen widget, homescreen banner — each channel has different attention rules and drop-off risks.)
- How does the offer address the user? Factual-informative ('15% off at Café Müller, 300m away') or emotional-situational ('Cold outside? Your cappuccino is waiting.')?
- What happens in the first 3 seconds? The offer must be understood without scrolling or deliberation. How does your UI achieve this?
- How does the offer end? Expiry, acceptance, or dismissal — each should feel intentional and leave the user experience intact.

Show the UX clearly in your demo and highlight how it addressed these four points

Data Sources and Hints

Your tool should be grounded in real or realistic data — not only synthetic proxies. You are encouraged to incorporate signal categories from the sources below.

CONTEXT & LOCATION	
OpenWeatherMap / DWD	Real-time and forecast weather data by city/location — core context trigger signal.
Eventbrite / Local event APIs	Local event calendars — city festivals, sports events, concerts — for demand spike detection.
Google Maps Platform / OSM	POI data, footfall signals, route density — for proximity and relevance scoring.
MERCHANT & TRANSACTION DATA	
Simulated Payone transaction feed	Simulate or stub Payone transaction density data per merchant — a core DSV asset for identifying quiet periods and triggering dynamic offers.
AI & GENERATIVE UI	
On-device SLMs (Phi-3, Gemma, etc.)	Small language models running on-device for GDPR-compliant local personalisation — only an abstract 'intent' signal reaches the server.
React Native / Flutter GenUI	Generative UI frameworks for streaming dynamically generated interface components — the offer widget is built at runtime, not retrieved from a template library.

What Makes a Strong Submission

Strong Submissions...	Weak Submissions...
— Show real context in action. Demonstrate the system responding to a concrete scenario (e.g. rain + low transaction volume) and generating a specific, plausible offer.	— Build a beautiful UI that shows static dummy offers with no real generative logic behind them.
— Design for 3-second comprehension. The offer must be understood instantly. Show how your UI achieves this — layout, language, hierarchy.	— Treat the merchant and customer side as an afterthought. The interface — even a mockup — is a required deliverable.
— Close the loop. Show the full journey: context detection → offer generation → display → accept/decline → simulated checkout. Even a partial but connected flow beats a polished stub.	— Over-engineer the AI stack and under-engineer the experience. This challenge is won in the interaction design, not the model architecture.
— Be honest about privacy. Address GDPR explicitly — how does your system protect user data? On-device inference, anonymisation, or consent flows all count.	— Ignore the merchant's perspective. A city wallet without happy merchants has no supply side — and no future.

Why This Matters

The decline of inner-city retail is a real structural threat — not only to local economies but to the regional business model of savings banks (Sparkassen) that are embedded in those same communities. Traditional loyalty programmes and static coupon books have failed to slow it. Meanwhile global e-commerce platforms have access to dynamic pricing, real-time demand signals, and algorithmic personalisation that local merchants simply cannot match.

DSV Gruppe — as part of the German Savings Banks Financial Group — sits at the intersection of payments infrastructure (Payone), merchant portals (S-Markt & Mehrwert), and regional banking relationships. This is a unique position from which to build something global e-commerce cannot replicate: an AI layer that knows the local context, respects privacy by design, and makes the merchant on the corner as responsive to demand as a marketplace algorithm.

Appendix — Contact & Additional Resources

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